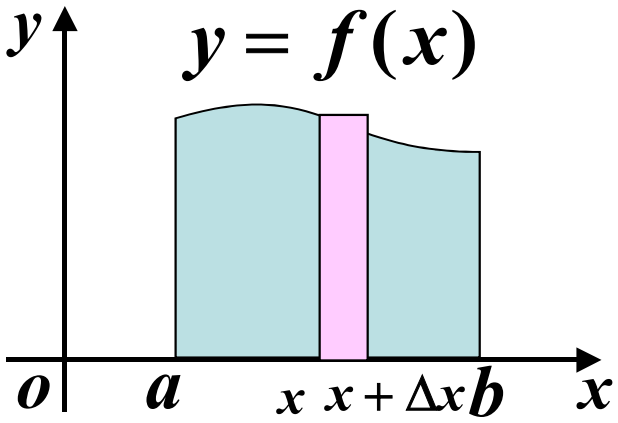
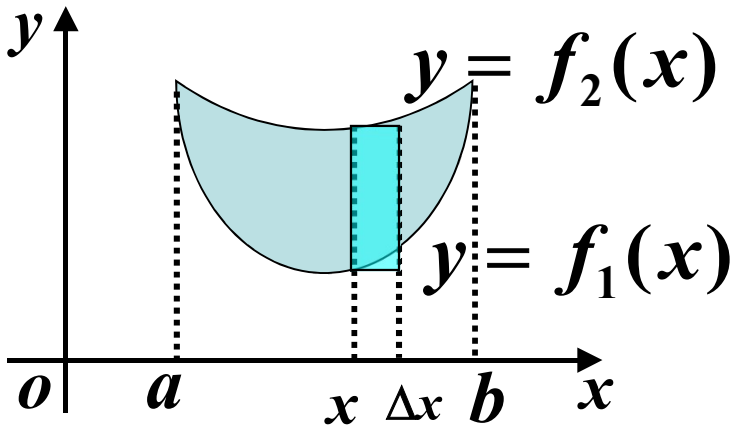
微元法求面积

一、直角坐标系情形



曲边梯形的面积

$$A=\int_a^b f(x)dx$$



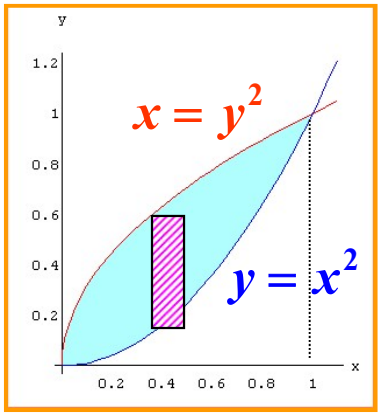
曲边梯形的面积

$$A=\int_a^b [f_2(x)-f_1(x)]dx$$

例 计算由两条抛物线 $y^2 = x$ 和 $y = x^2$ 所围成的图形的面积.

解 两曲线的交点

$$(0,0) \quad (1,1)$$



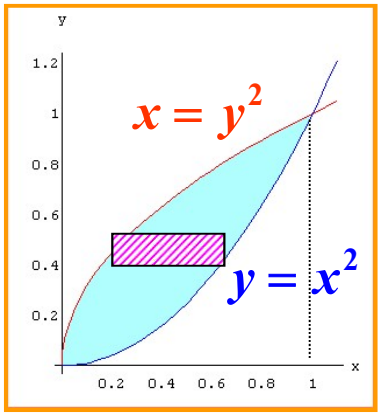
选 x 为积分变量 $x \in [0,1]$

面积元素 $dA = (\sqrt{x} - x^2)dx$

$$A = \int_0^1 (\sqrt{x} - x^2)dx = \left[\frac{2}{3}x^{\frac{3}{2}} - \frac{x^3}{3}\right]_0^1 = \frac{1}{3}.$$

解2 两曲线的交点

$$(0,0) \quad (1,1)$$



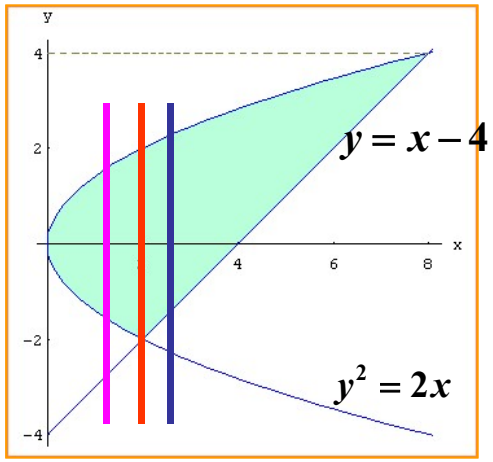
选 y 为积分变量 $y \in [0,1]$

面积元素 $dA = (\sqrt{y} - y^2)dy$

$$A = \int_0^1 (\sqrt{y} - y^2)dy = \frac{1}{3}.$$

例 计算由曲线$y^2=2x$和直线$y=x-4$所围成的图形的面积.

解 两曲线的交点



$$\begin{cases} y^2=2x \\ y=x-4 \end{cases} \Rightarrow (2,-2),(8,4).$$

选 y 为积分变量 $y\in[-2,4]$

$$dA=\left(y+4-\frac{y^2}{2}\right)dy \qquad A=\int_{-2}^{4}dA=18.$$

如果曲边梯形的曲边为参数方程 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}$

曲边梯形的面积 $A = \int_{t_1}^{t_2} \psi(t)\varphi'(t)dt.$

（其中t_1和t_2对应曲线起点与终点的参数值）

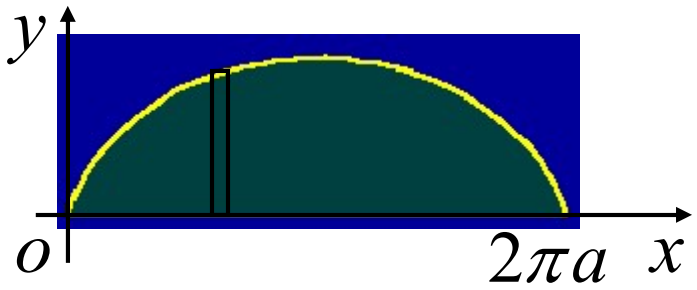
在$[t_1, t_2]$（或$[t_2, t_1]$）上$x = \varphi(t)$具有连续导数，

$y = \psi(t)$连续.

例. 求由摆线 $x = a(t - \sin t),\ y = a(1 - \cos t) \quad (a > 0)$ 的一拱与 x 轴所围平面图形的面积 .

解: $A = \int_0^{2\pi} a(1-\cos t)\cdot a(1-\cos t)\,\mathrm{d}t$

$$= a^2 \int_0^{2\pi} (1-\cos t)^2\,\mathrm{d}t$$



$$= 4a^2 \int_0^{2\pi} \sin^4 \frac{t}{2}\,\mathrm{d}t$$

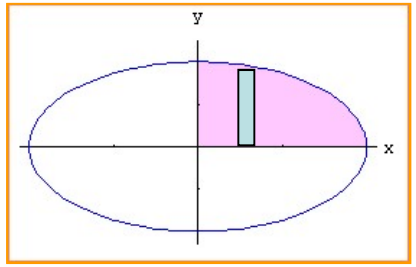
$$= 8a^2 \int_0^{\pi} \sin^4 u\,\mathrm{d}u \qquad (\text{令 } u = \frac{t}{2})$$

$$= 16a^2 \int_0^{\frac{\pi}{2}} \sin^4 u\,\mathrm{d}u$$

$$= 16a^2 \cdot \frac{3}{4} \cdot \frac{1}{2} \cdot \frac{\pi}{2} = 3\pi a^2$$

例 求椭圆$\dfrac{x^2}{a^2}+\dfrac{y^2}{b^2}=1$的面积.

解 椭圆的参数方程 $\begin{cases} x = a\cos t \\ y = b\sin t \end{cases}$

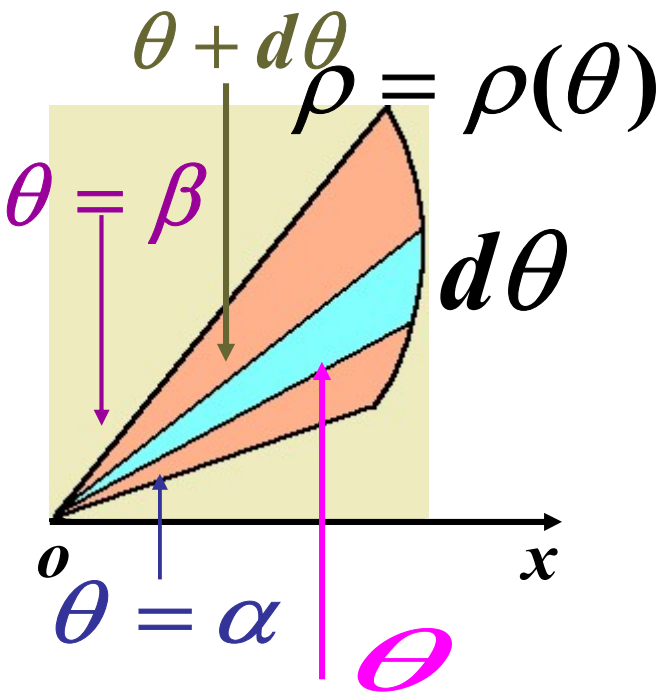


由对称性知总面积等于4倍第一象限部分面积.

$$A = 4\int_0^a y\,dx = 4\int_{\frac{\pi}{2}}^0 b\sin t\,d(a\cos t)$$

$$= 4ab\int_0^{\frac{\pi}{2}} \sin^2 t\,dt = \pi ab.$$

二、极坐标系情形



设由曲线 $\rho = \rho(\theta)$ 及射线 $\theta = \alpha$、$\theta = \beta$ 围成一曲边扇形，求其面积．这里，$\rho(\theta)$ 在 $[\alpha, \beta]$ 上连续，且 $\rho(\theta) \geq 0$．

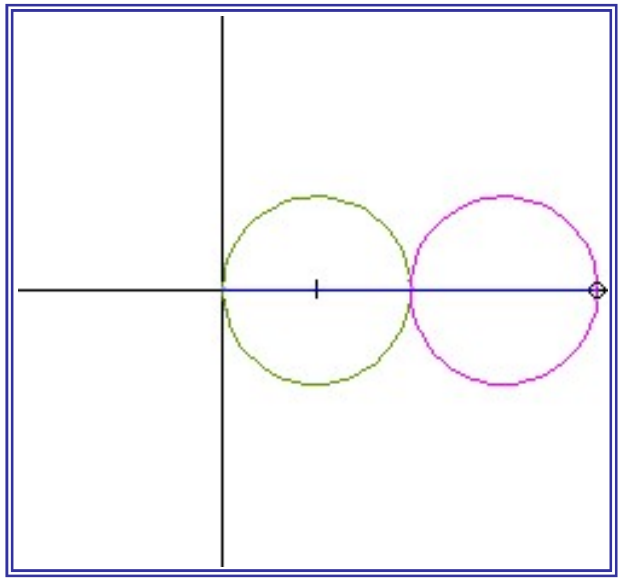
面积元素 $$dA = \frac{1}{2}[\rho(\theta)]^2 d\theta$$

曲边扇形的面积 $$A = \int_{\alpha}^{\beta} \frac{1}{2}[\rho(\theta)]^2 d\theta.$$

心形线(外摆线的一种)

$$x^2+y^2-ax=a\sqrt{x^2+y^2}$$

即 $\rho=a(1+\cos\theta)$



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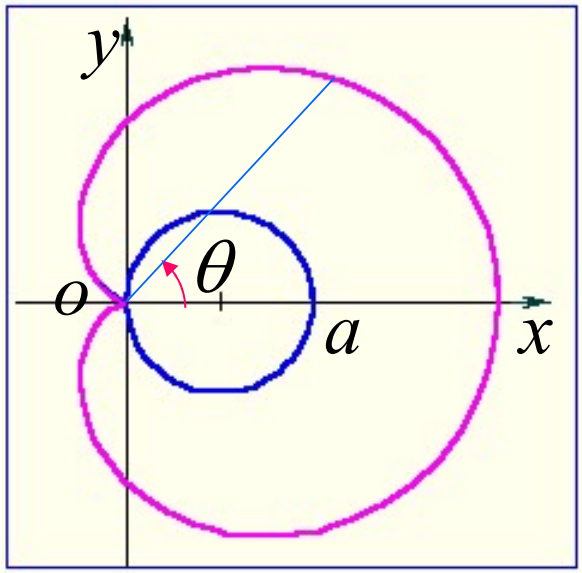
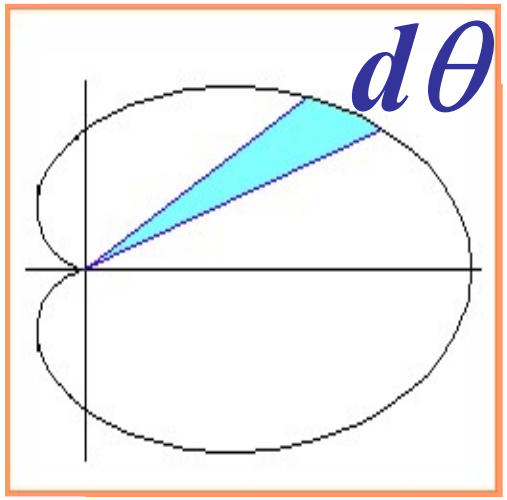
参数的几何意义



- 尖点: $(0,0)$
- 面积: $\frac{3}{2}\pi a^2$
- 弧长: $8a$

例 求心形线 $\rho = a(1+\cos\theta)$ 所围平面图形的面积$(a>0)$.



解 $dA = \frac{1}{2}a^2(1+\cos\theta)^2 d\theta$

利用对称性知

$$A = 2\cdot\frac{1}{2}a^2\int_0^{\pi}(1+\cos\theta)^2 d\theta$$

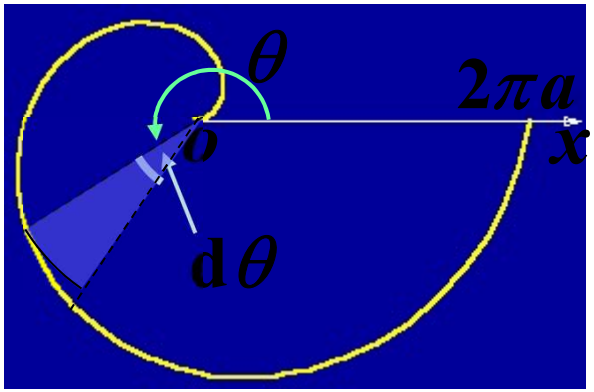
$$= \frac{3}{2}\pi a^2.$$

例. 计算阿基米德螺线 $\rho = a\theta$ 对应 θ 从 0 变到 2π 所围图形面积 .

解: $A = \int_0^{2\pi} \frac{1}{2}(a\theta)^2 \, \mathrm{d}\theta$

$$= \frac{a^2}{2}\left[\frac{1}{3}\theta^3\right]_0^{2\pi}$$

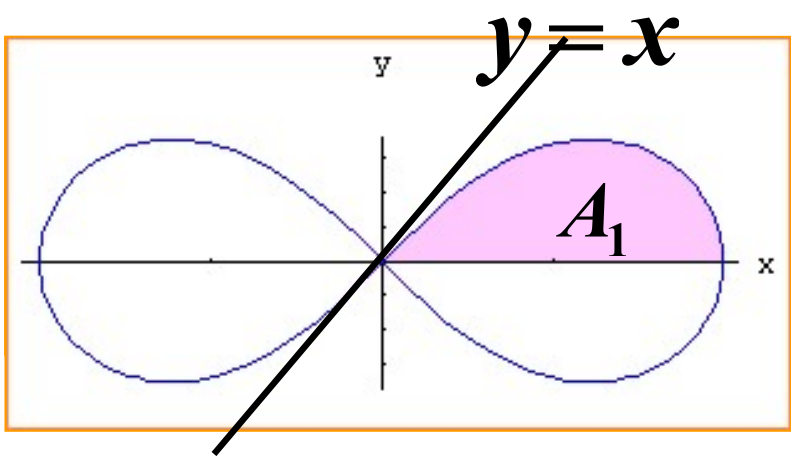
$$= \frac{4}{3}\pi^3 a^2$$



点击图片任意处
播放开始或暂停

例 求双纽线$\rho^2 = a^2\cos 2\theta$所围平面图形的面积.

$$\rho^2 = a^2\cos 2\theta$$



解 由对称性知总面积=4倍第一象限部分面积

$$A = 4A_1$$

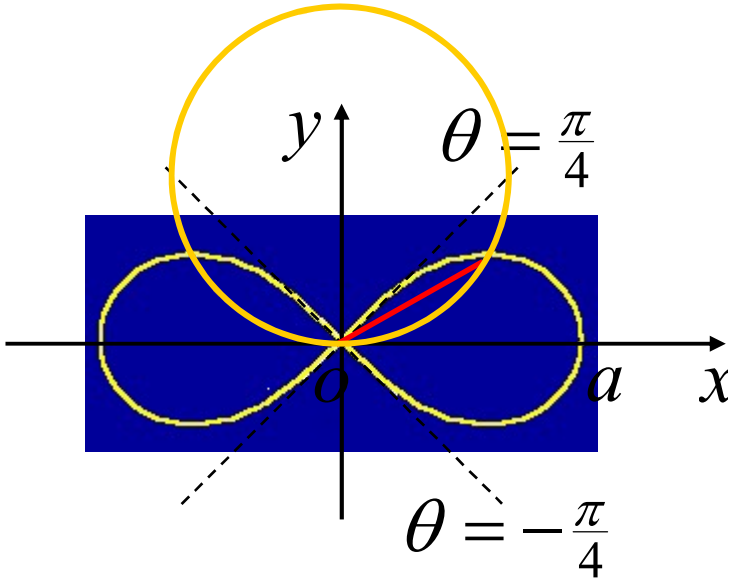
$$A = 4\int_0^{\frac{\pi}{4}} \frac{1}{2}a^2\cos 2\theta d\theta$$

$$= a^2.$$

思考: 用定积分表示该双纽线与所围公共部分的面积 .

$$\rho = a\sqrt{2}\sin\theta$$

答案: $$A = 2\left[\int_0^{\frac{\pi}{6}} a^2 \sin^2\theta \,\mathrm{d}\theta + \int_{\frac{\pi}{6}}^{\frac{\pi}{4}} \frac{1}{2} a^2 \cos 2\theta \,\mathrm{d}\theta\right]$$



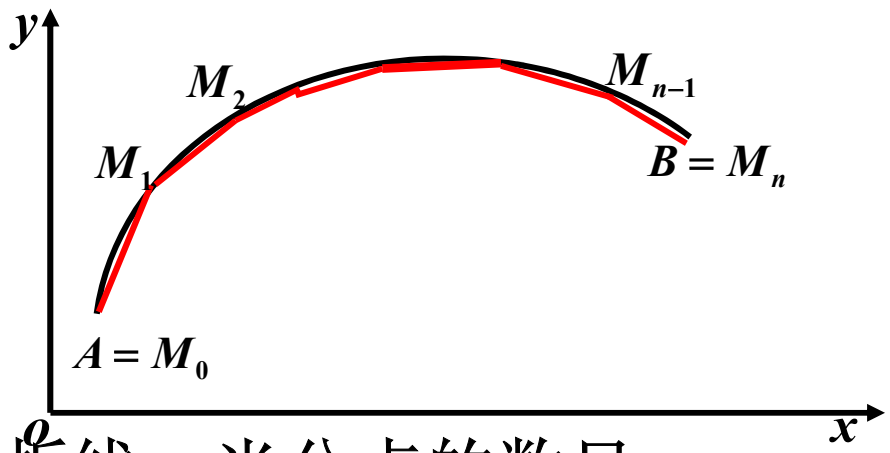
微元法求曲线的弧长

一、平面曲线弧长的概念

设A、B是曲线弧上的两个端点，在弧上插入分点

$$A = M_0, M_1, \cdots M_i,$$

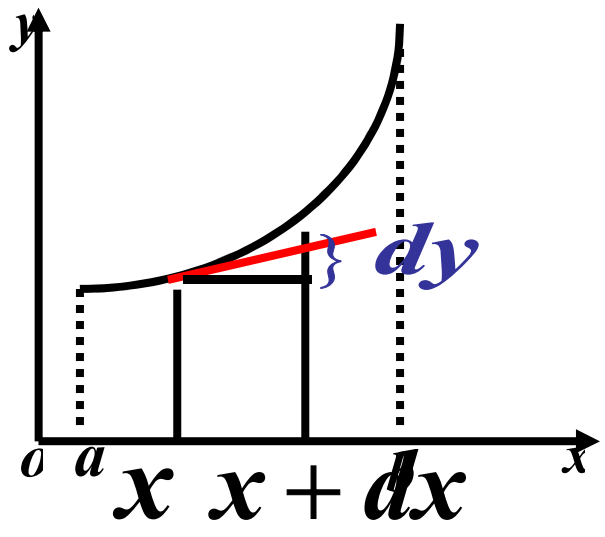
$$\cdots, M_{n-1}, M_n = B$$



并依次连接相邻分点得一内接折线，当分点的数目无限增加且每个小弧段都缩向一点时，

此折线的长$\sum_{i=1}^{n} |M_{i-1}M_i|$的极限存在，则称此极限为曲线弧$AB$的弧长.

二、直角坐标情形



设曲线弧为$y = f(x)$ $(a \le x \le b)$，其中$f(x)$在$[a,b]$上有一阶连续导数

取积分变量为x，在$[a,b]$上任取小区间$[x, x+dx]$，

以对应小切线段的长代替小弧段的长

小切线段的长 $\sqrt{(dx)^2 + (dy)^2} = \sqrt{1 + y'^2}dx$

弧长元素 $ds = \sqrt{1 + y'^2}dx$ 弧长$s = \int_a^b \sqrt{1 + y'^2}dx.$

三、参数方程情形

曲线弧为 $\begin{cases} x = \varphi(t) \\ y = \psi(t) \end{cases}, \qquad (\alpha \le t \le \beta)$

其中$\varphi(t), \psi(t)$在$[\alpha, \beta]$上具有连续导数.

$$ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{[\varphi'^2(t) + \psi'^2(t)](dt)^2}$$

$$= \sqrt{\varphi'^2(t) + \psi'^2(t)}dt$$

弧长

$$s = \int_\alpha^\beta \sqrt{\varphi'^2(t) + \psi'^2(t)}dt.$$

四、极坐标情形

曲线弧为 $\rho = \rho(\theta) \qquad (\alpha \le \theta \le \beta)$

其中 $\rho(\theta)$ 在$[\alpha, \beta]$上具有连续导数.

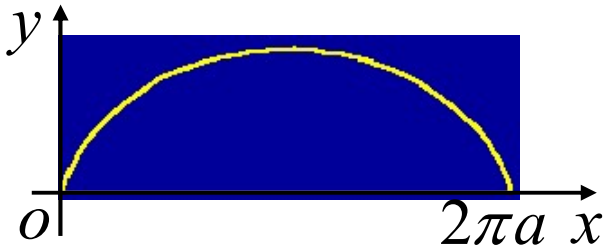
$$\because \begin{cases} x = \rho(\theta)\cos\theta \\ y = \rho(\theta)\sin\theta \end{cases} \qquad (\alpha \le \theta \le \beta)$$

$$\therefore ds = \sqrt{(dx)^2 + (dy)^2} = \sqrt{\rho^2(\theta) + \rho'^2(\theta)}d\theta,$$

弧长 $s = \int_\alpha^\beta \sqrt{\rho^2(\theta) + \rho'^2(\theta)}d\theta.$

例 计算摆线 $\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$ $(a > 0)$ 一拱 $(0 \le t \le 2\pi)$

的弧长 .



解: $\mathrm{d}s = \sqrt{(\frac{\mathrm{d}x}{\mathrm{d}t})^2 + (\frac{\mathrm{d}y}{\mathrm{d}t})^2}\,\mathrm{d}t$

$$= \sqrt{a^2(1-\cos t)^2 + a^2 \sin^2 t}\,\mathrm{d}t$$

$$= a\sqrt{2(1-\cos t)}\,\mathrm{d}t$$

$$= 2a \sin\frac{t}{2}\mathrm{d}t$$

$$\therefore \quad s = \int_0^{2\pi} 2a \sin\frac{t}{2}\mathrm{d}t = 2a\left[-2\cos\frac{t}{2}\right]_0^{2\pi} = 8a$$

例. 求连续曲线段 $y=\int_{-\frac{\pi}{2}}^{x}\sqrt{\cos t}\,\mathrm{d}t$ **的弧长.**

解: $\because\ \cos x\geq 0,\ \therefore\ -\frac{\pi}{2}\leq x\leq\frac{\pi}{2}$

$$s=\int_{-\frac{\pi}{2}}^{\frac{\pi}{2}}\sqrt{1+y'^2}\,\mathrm{d}x$$

$$=2\int_{0}^{\frac{\pi}{2}}\sqrt{1+(\sqrt{\cos x})^2}\,\mathrm{d}x$$

$$=2\int_{0}^{\frac{\pi}{2}}\sqrt{2}\cos\frac{x}{2}\mathrm{d}x$$

$$=2\sqrt{2}\left[2\sin\tfrac{x}{2}\right]_{0}^{\frac{\pi}{2}}$$

$$=4$$

五、小结

平面曲线弧长的概念

弧微分的概念

求弧长的公式
- 直角坐标系下
- 参数方程情形下
- 极坐标系下

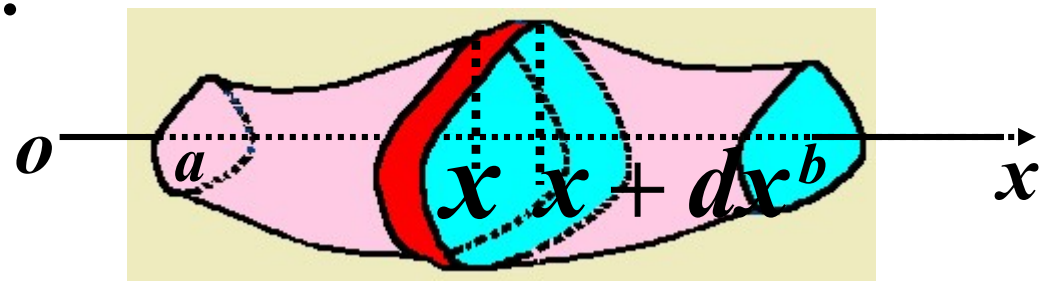
微元法求体积

一、平行截面面积为已知的立体的体积

如果一个立体不是旋转体，但却知道该立体上垂直于一定轴的各个截面面积，那么，这个立体的体积也可用定积分来计算.

$A(x)$表示过点

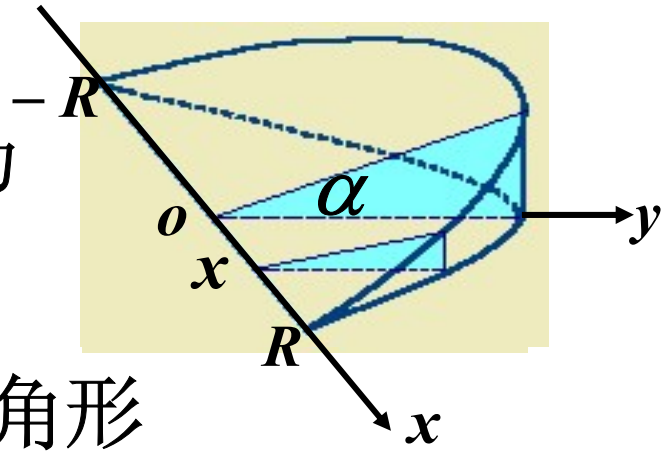
x且垂直于x轴



的截面面积，$A(x)$为x的已知连续函数

$$dV = A(x)dx,$$ 立体体积 $$V = \int_a^b A(x)dx.$$

例 一平面经过半径为R的圆柱体的底圆中心，并与底面交成角α，计算这平面截圆柱体所得立体的体积.



解 取坐标系如图 底圆方程为

$$x^2+y^2=R^2$$

垂直于x轴的截面为直角三角形

截面面积 $A(x)=\dfrac{1}{2}(R^2-x^2)\tan\alpha,$

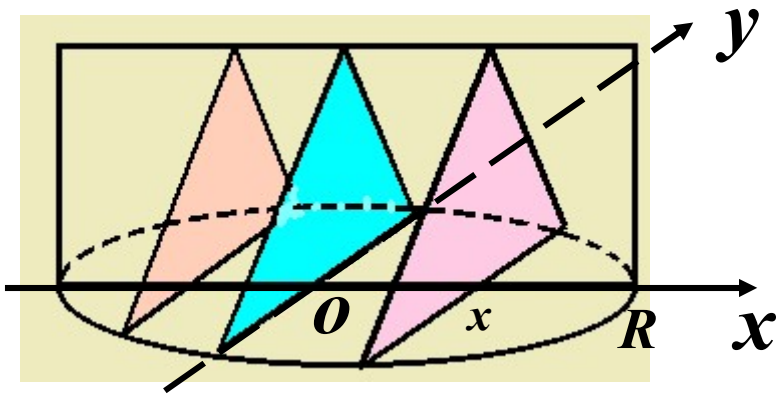
立体体积 $V=\dfrac{1}{2}\displaystyle\int_{-R}^{R}(R^2-x^2)\tan\alpha dx=\dfrac{2}{3}R^3\tan\alpha.$

例 求以半径为R的圆为底、平行且等于底圆直径的线段为顶、高为h的正劈锥体的体积.

解 取坐标系如图

底圆方程为

$$x^2 + y^2 = R^2,$$



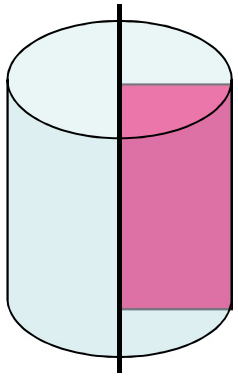
垂直于x轴的截面为等腰三角形

截面面积 $A(x) = h \cdot y = h\sqrt{R^2 - x^2}$

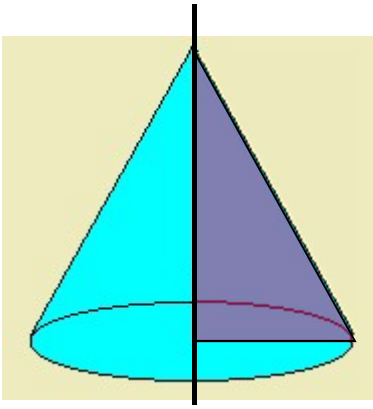
立体体积 $V = h\int_{-R}^{R}\sqrt{R^2 - x^2}dx = \frac{1}{2}\pi R^2 h.$

二、旋转体的体积

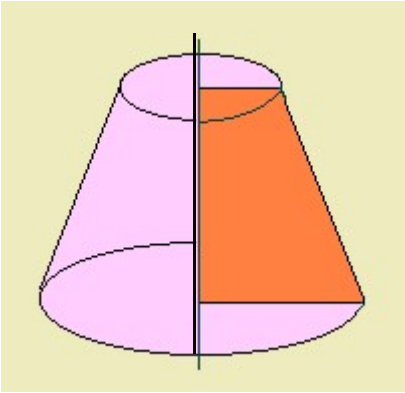
旋转体就是由一个平面图形绕这平面内一条直线旋转一周而成的立体．这直线叫做旋转轴．



圆柱



圆锥

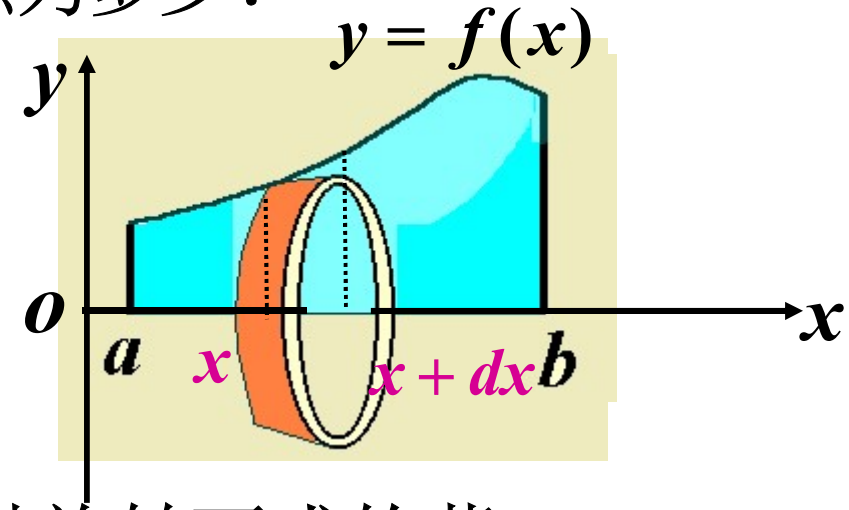


圆台

一般地，如果旋转体是由连续曲线 $y=f(x)$、直线 $x=a$、$x=b$ 及 x 轴所围成的曲边梯形绕 x 轴旋转一周而成的立体，体积为多少？

取积分变量为 x，$x\in[a,b]$

在 $[a,b]$ 上任取小区间 $[x,x+dx]$，

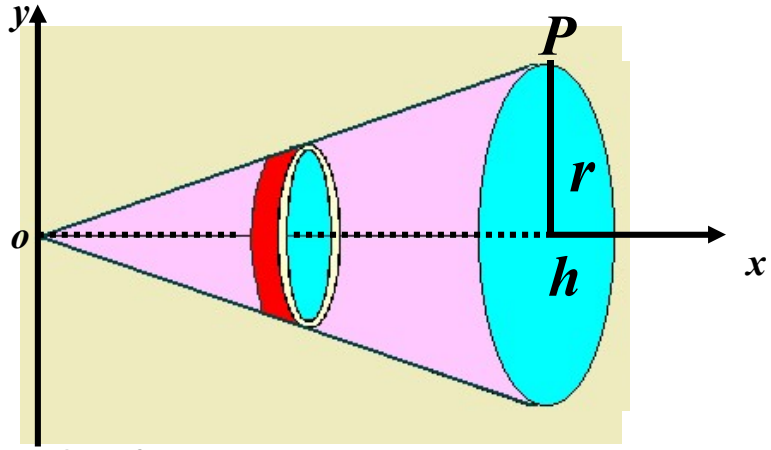


取以 dx 为底的窄边梯形绕 x 轴旋转而成的薄片的体积为体积元素，　$dV=\pi[f(x)]^2dx$

旋转体的体积为

$$V=\int_a^b \pi[f(x)]^2dx$$

例 连接坐标原点O及点$P(h,r)$的直线、直线$x=h$及x轴围成一个直角三角形. 将它绕x轴旋转构成一个底半径为r、高为h的圆锥体，计算圆锥体的体积.



解 直线 OP方程为

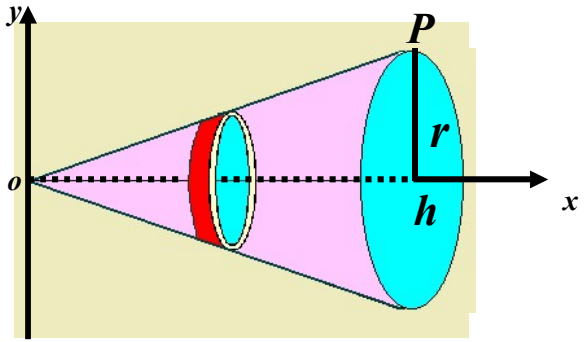
$$y=\frac{r}{h}x$$

取积分变量为x， $x\in[0,h]$

在$[0,h]$上任取小区间$[x,x+dx]$，

以dx为底的窄边梯形绕x轴旋转而成的薄片的体积为

$$dV = \pi\left[\frac{r}{h}x\right]^2 dx$$



圆锥体的体积

$$V = \int_0^h \pi\left(\frac{r}{h}x\right)^2 dx = \frac{\pi r^2}{h^2}\left[\frac{x^3}{3}\right]_0^h = \frac{\pi h r^2}{3}.$$

星形线 $x = a\cos^3 t,\ y = a\sin^3 t$

星形线是内摆线的一种.(当小圆在圆内沿圆周滚动时, 小圆上的定点的轨迹为是内摆线)

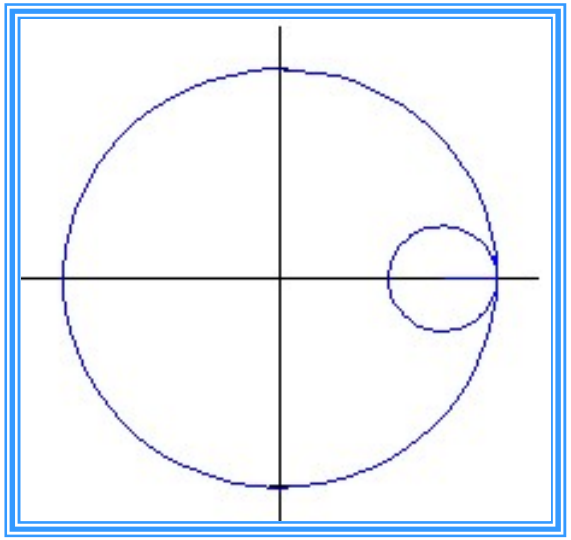
参数的几何意义

大圆半径

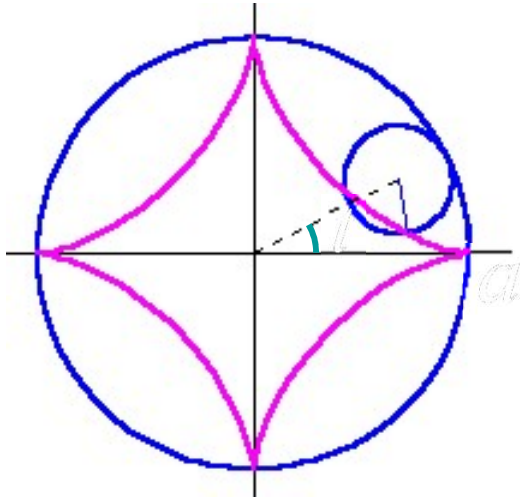
$R = a$

小圆半径

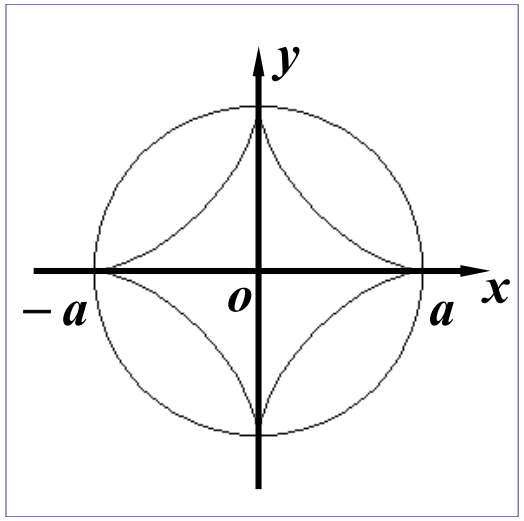
$$r = \frac{a}{4}$$



点击图片任意处
播放开始或暂停



例 求星形线$x^{\frac{2}{3}}+y^{\frac{2}{3}}=a^{\frac{2}{3}}(a>0)$绕$x$轴旋转构成旋转体的体积.



解 $\because y^{\frac{2}{3}}=a^{\frac{2}{3}}-x^{\frac{2}{3}}$,

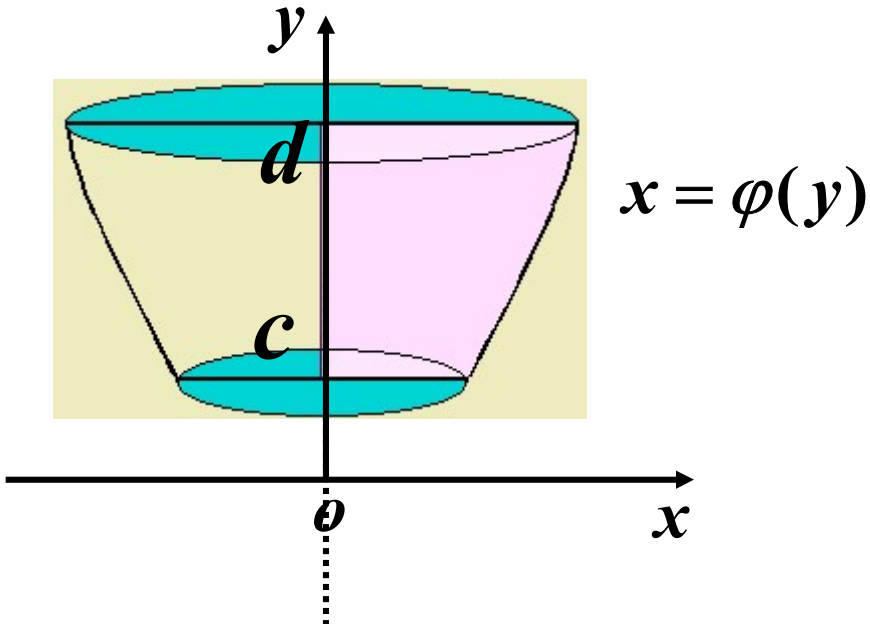
$$\therefore y^2=\left(a^{\frac{2}{3}}-x^{\frac{2}{3}}\right)^3 \quad x\in[-a,a]$$

旋转体的体积

$$V=\int_{-a}^{a}\pi\left(a^{\frac{2}{3}}-x^{\frac{2}{3}}\right)^3dx=\frac{32}{105}\pi a^3.$$

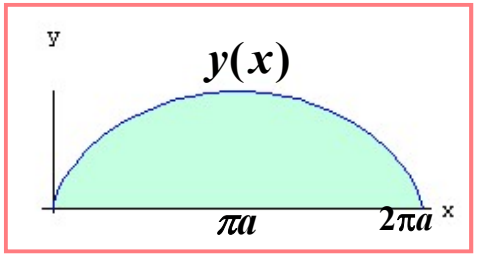
类似地，如果旋转体是由连续曲线 $x=\varphi(y)$、直线 $y=c$、$y=d$ 及 y 轴所围成的曲边梯形绕 y 轴旋转一周而成的立体，体积为

$$V=\int_c^d \pi[\varphi(y)]^2 dy$$



例 求摆线$x=a(t-\sin t)$，$y=a(1-\cos t)$的一拱与$y=0$所围成的图形分别绕x轴、y轴旋转构成旋转体的体积.

解 绕x轴旋转的旋转体体积

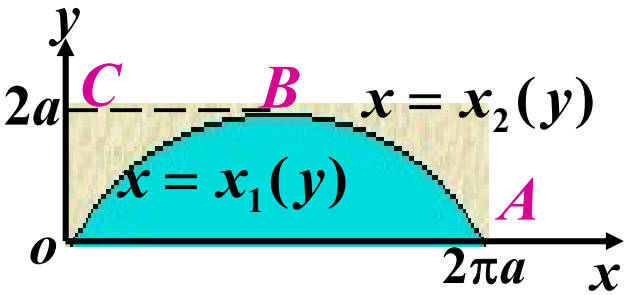


$$V_x=\int_0^{2\pi a}\pi y^2(x)dx$$

$$=\pi\int_0^{2\pi}a^2(1-\cos t)^2\cdot a(1-\cos t)dt$$

$$=\pi a^3\int_0^{2\pi}(1-3\cos t+3\cos^2 t-\cos^3 t)dt=5\pi^2a^3.$$

绕y轴旋转的旋转体体积



可看作平面图$OABC$与OBC

分别绕y轴旋转构成旋转体的体积之差.

$$V_y = \int_0^{2a} \pi x_2^2(y)dy - \int_0^{2a} \pi x_1^2(y)dy$$

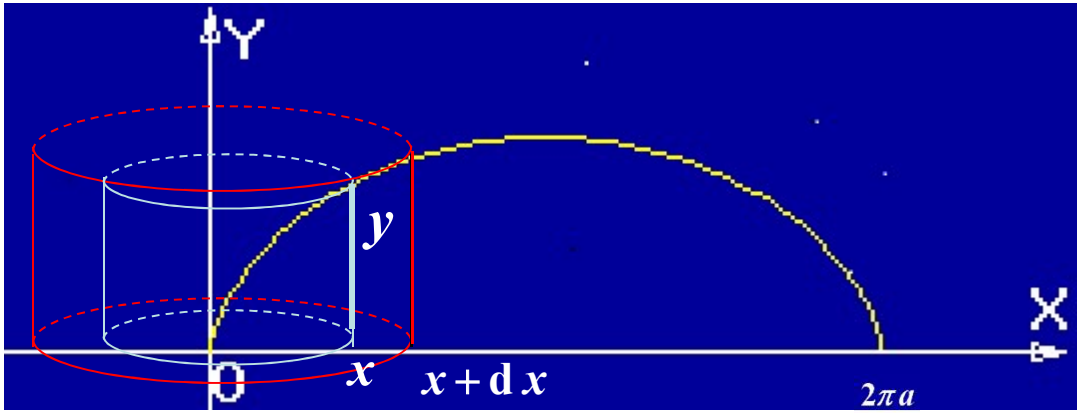
$$= \pi\int_{2\pi}^{\pi} a^2(t-\sin t)^2 \cdot a\sin t dt$$

$$- \pi\int_0^{\pi} a^2(t-\sin t)^2 \cdot a\sin t dt$$

$$= \pi a^3 \int_0^{2\pi} (t-\sin t)^2 \sin t dt = 6\pi^3 a^3.$$

$$\begin{cases} x = a(t - \sin t) \\ y = a(1 - \cos t) \end{cases}$$

说明: V_y **也可按柱壳法求出**



柱面面积 $2\pi x \cdot y$ **柱壳体积** $2\pi xy \cdot \mathrm{d}x$

$$V_y = 2\pi \int_0^{2\pi a} xy\,\mathrm{d}x$$

$$= 2\pi \int_0^{2\pi} a(t - \sin t) \cdot a^2 (1 - \cos t)^2 \,\mathrm{d}t$$

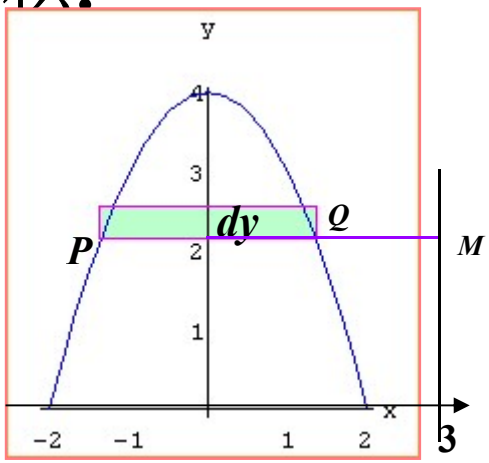
补充 如果旋转体是由连续曲线$y=f(x)$、直线$x=a$、$x=b$及x轴所围成的曲边梯形绕y轴旋转一周而成的立体，体积为

$$V_y = 2\pi\int_a^b x\,|f(x)|\,dx$$

利用这个公式，可知上例中

$$\begin{aligned} V_y &= 2\pi\int_0^{2\pi a} x\,|f(x)|\,dx \\ &= 2\pi\int_0^{2\pi} a(t-\sin t)\cdot a(1-\cos t)\,d[a(t-\sin t)] \\ &= 2\pi a^3\int_0^{2\pi}(t-\sin t)(1-\cos t)^2\,dt = 6\pi^3 a^3. \end{aligned}$$

例 求由曲线$y=4-x^2$及$y=0$所围成的图形绕直线$x=3$旋转构成旋转体的体积.



解 取积分变量为y, $y\in[0,4]$

体积元素为

$$dV=[\pi\overline{PM}^2-\pi\overline{QM}^2]dy$$

$$=[\pi(3+\sqrt{4-y})^2-\pi(3-\sqrt{4-y})^2]dy$$

$$=12\pi\sqrt{4-y}dy,$$

$$\therefore V=12\pi\int_0^4\sqrt{4-y}dy=64\pi.$$